KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF AGRICULTURAL AND BIOSYSTEMS ENGINEERING



AE 153 – Agricultural Meteorology & Climatology (2,1,2)

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Course Content

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AE 153 – Agricultural Meteorology & Climatology (2,1,2)

Importance and practical applications of agro-meteorology measurements and climatological data processing and analysis.

Course Goals

This course has four specific goals:

i. To introduce students to solar radiation and its relation with the atmosphere

ii. To introduce student to agro-meteorology measurement and its practical applications

iii. To develop analytical skills relevant to climate data processing and analysis

iv. To introduce students to climate variability and climate change

Reference Material

(1) Soil and water conservation engineering by G. O. Schwab

(2) Hydrology in practice by E. M. Shaw

(3) Agrometeorology: principles and applications of climate studies in agriculture by H S Mavi and G. J. Tupper

Course Assessment

i. Quizzes

ii. One (1) mid-semester test

iii. Two (2) project or practical assignment

iv. One (1) End-of-semester examination

AGROMETEOROLOGY AND CLIMATOLOGY

Weather: The state of the atmosphere at a given time and place, with respect to variables such as temperature, moisture, wind velocity, and barometric pressure.

Climate: Is the statistics of temperature, humidity, atmospheric pressure, wind, rainfall and other meteorological elements in a given region over long periods of time (usually 30 years). Climate can be contrasted to weather, which is the present condition of these same elements and their variations over periods of up to two weeks.

The climate of a location is affected by its latitude, terrain, vegetation and altitude nearby water bodies and their currents. Climate can be classified according to the average and the typical ranges of different variables, most commonly using temperature and precipitation. The most commonly used classification scheme was originally developed by Köppen.

Meteorology: The scientific study of the atmosphere and atmospheric conditions, especially as they relate to weather and weather forecasting.

Agrometeorology: The branch of meteorology that studies the effect of weather and climate on agriculture. It examines the effects and impacts of weather and climate on crops, rangeland, livestock, and various agricultural operations. Agricultural meteorologists collect, analyse and interpret weather and climate data needed to understand the interactions between vegetation and animals and their atmospheric environments. The climatic information developed by agricultural meteorologists is valuable in making proper decisions for managing resources consumed by agriculture, for optimizing agricultural production, and for adopting farming practices to minimize any adverse effects of agriculture on the environment. Such information is vital to ensure the economic and environmental sustainability of agriculture now and in the future. Agricultural meteorologists also quantify, evaluate, and provide information on the impact and consequences of climate variability and change on agriculture. Increasingly, agricultural meteorologists assist policy makers in developing strategies to deal with climatic events such as floods, rain storms, or droughts and climatic changes such as global warming and climate variability.

SOLAR RADIATION AND THE ATMOSPHERE

SOLAR RADIATION

Solar radiation (insolation) is the primary energy source for the earth surface. The amount of insolation reaching the earth surface assuming there is atmosphere is determined by the

1. **Solar output:** the amount of energy emitted by the sun towards the earth surface. Almost half of the energy emitted by the sun is in the visible light range (0.4-0.7 μm) with the rest as ultraviolet or infrared waves. The total solar radiation received per unit time on a unit area placed at right angles to the sun's rays at the earth's mean distance from the sun is referred to as the solar constant (1.39 KW m^{-2} or 1.99 cal cm^{-2} min^{-1}).
2. **Distance from the sun:** As a result of the eccentric orbit of the earth, the distance of any point on it from the sun changes continuously. Therefore the solar constant varies accordingly.
3. **Altitude of the sun:** Solar radiation received varies with altitude of the sun that is determine by the latitudinal position, season and time of the day
4. **Length of day**: the amount of solar radiation reaching any point on the earth depends on day length, which is influenced by the latitude and season

Importance of radiation

1. Primary source of heat to the earth and its atmosphere
2. Makes the planet habitable
3. Powers the dynamics of the atmosphere (e.g. the wind systems, hydrological cycle and therefore the weather)
4. Earth's seasons are controlled by the changes in the duration and intensity of solar radiation
5. Plant growth processes through photosynthesis, respiration and warming up the earth's surface

Characteristics of energy from the sun

1. The solar heat energy travels through space in the form of electromagnetic waves enabling the transfer of heat through radiation.
2. All bodies emit radiation and that the wavelength and energy characteristics of that radiation are determined by the body's temperature (Wien's Displacement Law)
3. The radiant energy drops with the square of distance from the radiating body (Inverse square law)
4. Radiation goes through a transformation when it encounters other objects (solid, gas or liquid)

Earth-Sun relationships: revolution, rotation, and tilt

Revolution around sun (i.e. shape of orbit): → seasons and climate variability

Revolution around sun + tilt → sometimes in the year Northern Hermisphere gets more sun, sometimes Southern Hermisphere gets more. Seasons and day length are the result of revolution and tilt of the earth which influence the variation in amount of incoming energy from the sun.

Rotates around its axis at north and south poles: → wind patterns, day and night pattern, ocean currents.

Tilt: variation in day length, tropical and arctic boundaries

Electromagnetic radiation

Radiation is electromagnetic energy travelling in the form of waves at the speed of light 300,000 kilometers per second. It can also be described as a stream of energetic particles without mass, known as **photons**. Electromagnetic radiation is most often classified according to its **wavelength**; each photon has a certain wavelength and the amount of energy carried by that photon depends on its wavelength. Photons with longer wavelengths carry lower energy, while photons with shorter wavelengths carry higher energy. Typical wavelengths are very small - about one millionth of a

meter; this unit is called the micrometer or micron. The different types of radiation over the entire range of wavelengths, from a fraction of a micrometer to hundreds of meters, is referred to as the **electromagnetic spectrum**.

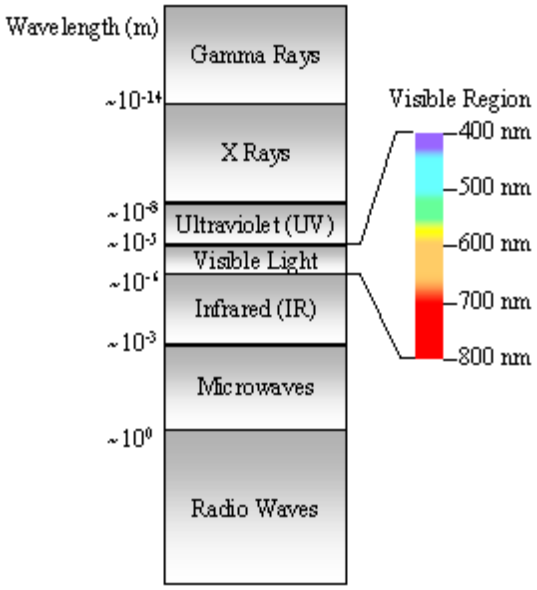


Figure 1: The electromagnetic spectrum

For the earth and atmosphere, we are most interested in the following wavelength regions of the electromagnetic spectrum:

- **Ultraviolet (UV)**: wavelengths from about 0.1 to 0.4 micrometers
- **Visible**: wavelengths from 0.4 (violet) to 0.7 (red) micrometers
- **Infrared (IR)**: wavelengths from about 0.7 to 10 micrometers

The amount of radiation emitted by a body depends on its temperature, and is proportional to T^4. This relation shows that as the temperature of the object increases, the amount of radiation emitted increases very rapidly. The emitted radiation will travel at the speed of light until it is absorbed by another body. The absorbing medium can be gas, liquid, or solid. Radiation does not require a medium to pass through.

Stefan-Boltzman Law

The total energy (radiant) emitted by a body surface (irradiance E) of an object, regardless of the wavelengths can be calculated as

$$E = \varepsilon \sigma T^4$$

where: T = temperature of the radiating object in K = °C + 273

σ = Stefan-Boltzman constant i.e. $5.67 \times 10^{-8}\ W/m^2\ K^4$ or 8.13×10^{-11} cal cm^{-2} min^{-1} deg^{-4}K. ε = emissivity that determines how well a surface can absorb or emit energy ($0 < \varepsilon \leq 1$) Note for a black body $\varepsilon_b = 1$

Calculate surface irradiance E of the sun for a surface temperature of 5700 K and emissivity of 1

Wein's displacement law

The sun emit somehow like a black body at the temperature of 6000°K. For any given body the emission reaches a max at a particular wavelength (λ_{max}) at a given temperature, and that λ decreases afterward as the temperature increases. In accordance with **Wein's displacement law** with λ measured in microns ($\mu = 10^{-6}$ m) and T in °K

$$\lambda_{max} \text{ proportional to } T^{-1}$$

$$\lambda_{max} T = 2900$$

The two prominent characters of the wave are the wavelength (λ) and frequency (ν). The wavelength is the distance between crest to crest on the wave. The frequency is related to wavelength by the following:

$$\lambda = \frac{c}{\nu}$$

where c is the speed of light, approximately equal to 3×10^8 m/s in vacuum. The wavelength is measured in units of length, and the frequency is given in cycles per second (*hertz*, Hz).

Inverse square Law

The Inverse Square Law is used to calculate the decrease in radiation intensity due to an increase in distance from the radiation source.

Inverse Square Law:

$$\frac{4\pi R_{SUN}^2}{4\pi r_{ES}^2} = \frac{I_{EARTH}}{E_{SUN}}$$

I_{EARTH} = Irradiance at the surface of any planetary object e.g. Earth
E_{SUN} = Irradiance at the surface of the object (Sun)

R= 6.96 x 10^5 km = Radius of the Sun ($4\pi R^2$ = surface area of the object)
$\mathbf{r_{ES}}$ = 1.5 x 10^8 km =Average Sun-Earth Distance ($4\pi r_{ES}^2$ = surface area of the outer sphere)

In order to calculate the solar constant the following equation is used:

$$I_{earth} = E_{SUN}\left(\frac{R_{SUN}}{r_{EARTH}}\right)^2$$

I_{EARTH} also known as Solar constant

Atmospheric effects on solar radiation

The atmosphere acts as a filter in both directions serving as protective shield of the earth from incoming shortwave solar radiation and also loss of heat. While passing through the atmosphere, the incoming solar radiation (*insolation*) is

partly: 1- Absorbed, 2- Reflected, 3- Scattered, and 4- Transmitted.

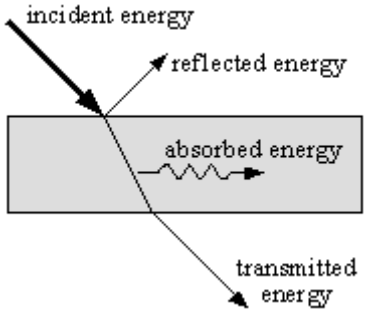


Figure 2: Atmospheric effect on insolation

Absorption: Atmospheric gases, particulates, and droplets all reduce the intensity of insolation by absorption. The gases in the atmosphere are not equally effective at absorbing sunlight.

Reflection: - The reflection of energy is a process whereby radiation making contact with some material is simply redirected away from the surface without being absorbed

The percentage of visible light reflected by an object is called albedo (reflectivity).

Albedo = 1 -> all is reflected

Table 1. Example of comparative Albedo levels for different surfaces

High	**Low**
light colors	dark colors
smooth	rough
snow	vegetation, asphalt, concrete

The reflectivity or albedo of the Earth's surface varies with the type of material that covers it.

Table 2. Albedo of surfaces

Surface	**Reflectivity/Albedo**
Fresh snow	0.75 - 0.9
Old Snow	0.4 - 0.7
grass	0.17 - 0.28
maize	0.18 - 0.22
Rain forest	0.12
Wet dark soil	0.08
Dry dark soil	0.13
Dry sand	0.35
Earth(average)	0.30
water	0.10
clouds	0.5 – 0.9

Scattering: Scattering is the process whereby a beam of radiation is broken down into many weaker rays redirected in other directions (diffuse scattering), without change in wavelength and there is no heating. Usually, more energy is scattered in the forward direction than is backscattered. There are two main types of scattering i.e. **Rayleigh and Mie scattering** depending on the wavelength of incoming radiation, direction (backward and or forward) of scattered radiation and size of scattering particles (agents).

Transmission: - Unmodified solar radiation, traveling through the vacuum of outer space and reaching the top of Earth's atmosphere.

- Absorption coefficient or absorptivity (α): Is the fraction of incident radiation intensity absorbed by a body
- Reflection coefficient (ρ): Is the fraction of incident radiation intensity that is reflected
- Transmission coefficient (τ): Is the fraction of incident radiation intensity that is transmitted

From the principle of energy conservation; $\alpha + \rho + \tau = 1$

Black body: Is a body that absorbs all the energy that falls on it. Therefore for a black body the absorptivity $\alpha_b = 1$

ATMOSPHERE

The atmosphere surrounds the earth. It is made up of a body of air or gasses that protects the planet and supports life and climate processes.

There are three main factors that directly influence the energy balance of the earth and it's temperature:

The total energy influx, which depends on the earth's distance from the sun and on solar activity

1. The chemical composition of the atmosphere
2. Albedo, the ability of the earth's surface to reflect light.

Composition of the atmosphere

The atmosphere is the body of air which surrounds our planet. Most of our atmosphere is located close to the earth's surface where it is most dense.

Table 3. Atmospheric constituents

Constituent	Relative percentage of dry air (%)
Nitrogen	78.08
Oxygen	20.95
Argon	0.934
Carbon dioxide	0.036
Neon	0.0018
Helium	0.0005
Methane	0.00017
Hydrogen	0.00005
Nitrous	0.00003
Ozone	0.000004

** In addition, water vapor is variable but typically makes up about 1-4% of the atmosphere.

Layers of the Atmosphere

There are five distinct layers of the earth's atmosphere:

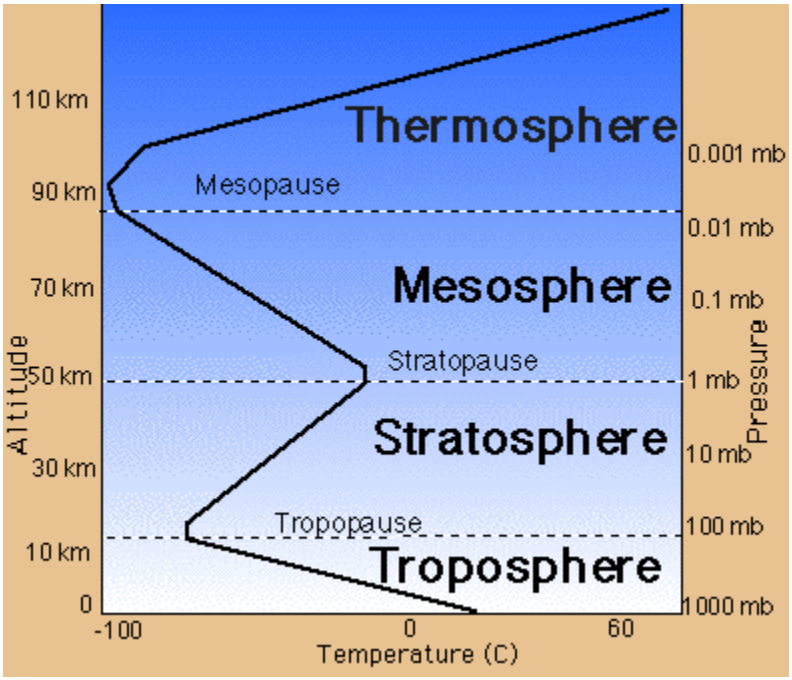


Figure 3: Variation of temperature through the layers of the atmosphere

Troposphere: The layer of the atmosphere closest to the earth is the troposphere. This layer is where weather occurs. It begins at the surface of the earth and extends out to about 6 -19 km. The temperature of the troposphere decreases with height. This layer is known as the lower atmosphere.

a. Vertical mixing of air and water vapor generates "weather."

b. Greenhouse gases, primarily H_2O, CO_2 and CH_4, regulate surface temperatures.

Stratosphere: Above the troposphere is the stratosphere, which extends to about 48-56 km above the earth's surface. Temperature rises within the stratosphere but still remains well below freezing. Photochemical reactions produce ozone.

Mesosphere: From about 56 to 80 km above the surface of the earth lies the mesosphere, where the air is especially thin and molecules are great distances apart.

Thermosphere: The thermosphere rises several hundred kilometres above the earth's surface, from 80 km up to about 644 km. It is also known as the upper atmosphere. Oxygen molecules absorb short-wave solar radiation, which results in heating.

Exosphere: Extending from the top of the thermosphere to 10,000 km above the earth is the exosphere. This layer has very few atmospheric molecules, which can escape into space.

The role of ozone

Ozone plays a key role in sustaining life on earth. It absorbs UV radiation that is harmful to biological molecules (such as DNA, proteins, other nucleic acids). Yet ozone forms only a minute fraction of the total atmosphere. Ozone is now being destroyed by the increasing concentration of man-made chemicals such as CFCs (chlorofluorocarbons) and nitrogen oxides such as NO_x and N_2O.in the upper atmosphere.

Earth's atmosphere effect on energy received at the Earth's surface

Sensible heat: Out of the 45% (units) of solar energy received at the Earth's surface, 8 units are used to warm the air through conduction and convection processes. The energy transferred in these processes is called sensible heat (increase in the air temperature that can be sensed). - **Latent heat**: 21 units of solar energy received at the Earth's surface, are used to evaporate water (change from liquid to vapor phase without change in temperature). The energy used for this process is hidden (latent) in the air and is released during condensation (change from vapor to liquid, such as precipitation).

The rest of the energy absorbed by the Earth's surface (45 - 8 - 21 = 16 units) and used to warm up the surface.

Earth's surface radiates energy almost completely in the *longwave 3 - 30 μm* (predominantly thermal infrared) portion of the spectrum. Longwave radiation emitted by the Earth's surface is largely absorbed by the atmosphere.

Net radiation

Net radiation is the balance between shortwave and longwave radiation, expressed

$$R_n = R_{si} - R_{so} + R_{li} - R_{lo}$$

Where:

- R_n is the net radiation,
- R_{si} is the incoming solar (shortwave) radiation,
- R_{so} is the reflected (outgoing) solar radiation,
- R_{li} is the incoming longwave (atmospheric) radiation,
- R_{lo} is the outgoing longwave (terrestrial) radiation.

All terms are expressed in W m^{-2}.

AGRO-METEOROLOGY MEASUREMENTS

Weather occurs in the part of the earth's atmosphere called the troposphere. This is the layer which is closest to the earth's surface, or the actual sky that we can see. The weather is the condition of the atmosphere (or changes in the atmosphere) at a particular time and place. The weather is shaped by 4 main factors: the air, the sun, water and the earth's movement. Weather instruments are all focused on one primary task of measuring atmospheric conditions. Meteorologists are interested in all of the effects that cause different types of weather, including: heat, pressure, moisture and wind.

Table 4. Key weather variables, units and equipment used for their measurement

Weather variable	**Units**	**Equipment**
Rainfall	mm	Standard Raingauge, Automatic Raingauge (continuous record of rainfall)
Wind run/speed	m/s	Cup Anemometer
Wind direction	°	wind vane
Sunshine	hour	Stoke sunshine recorder
Evaporation	mm	Evaporation Pan, Piche Evaporimeter
Dry & Wet bulb temperature	°C	Dry & Wet bulb Thermometer
Temperature & Humidity	°C and %	Thermohygrograph
Soil temperature	°C	Soil Thermometers
Atmospheric pressure	mbar	Barometer
Solar-radiation	W/m²	Radiometer
Atmospheric relative humidity	%	Psychrometer, hygrometer

Stevenson screen is a meteorological screen used to shield weather instruments against precipitation and direct heat radiation from outside sources while still allowing air to circulate freely around them. Stevenson screen houses thermometers (maximum, minimum dry bulb-and wet bulb-temperature), a hygrometer, a psychrometer, a barometer and a thermograph four types of thermometers. It is usually placed at a distance of about 2 m from the ground.



Figure 4: Stevenson screen

Precipitation

Precipitation types: Precipitation may be in the form of falling water droplet such as rain, drizzle (uniform precipitation with droplet sizes less that 0.5 mm) or frozen water particles such snow, sleet (raindrops fall through below freezing temperature air) and hail. Precipitation may also be in the form of dew, formed from the direct condensation and absorption from the atmosphere.

There are three primary steps for formation of precipitation:

A. Creation of saturation conditions in the atmosphere (typically when there is lifting of air mass)

B. Phase change from vapour to liquid (cloud formation)

C. Growth of droplets to appreciable size that will be able to overcome upward velocity

A. **Air Mass movement**: The characteristics of the air mass are based on where it originated and is subsequently so named. For example: Tropical continental (Tc) (hot and dry wind from deserts in low latitude eg. Sahara) and Tropical maritime ™ air mass (warm and moist from sub-tropical oceans)

Lifting Mechanisms

A. Front: the boundary between two adjacent air masses of different temperature and moisture content.

1. **Warm Front**: Warm air advancing on colder air

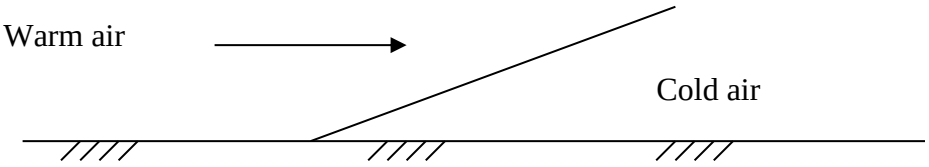


Figure 5: Warm air front

The convective rise (slope) of warm air is mild, with associated precipitation events having longer duration, lower intensity, and less energy resulting in gentler storms.

Durations: 3 – 10 hrs; Intensities: 1.25 – 5 mm/hr

2. **Cold front**: Cold air advancing on warm air

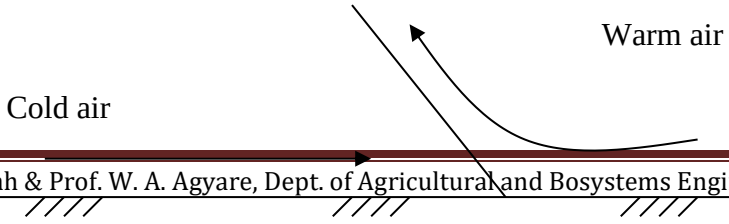


Figure 6: Cold air front

The convective rise (slope) is much steeper and can be nearly vertical. The associated precipitation events tend to have shorter duration, higher intensity, and more energy resulting in stronger more severe storms (the air movement is much faster).

Durations: 2 – 4 hrs; Intensities: 2.5 – 13.0 mm/hr

3. Cyclonic Events

a. ***Tropical Cyclones*:** These are non-frontal events but convective and generated by rising of warm-moist air masses. Typical requirements:

1. Intense sunshine (typical of ocean)
2. High humidity and temperature (typical of ocean)
3. Low surface friction (typical of ocean)

b. ***Monsoons*:** 180° reversal in prevailing wind direction with season. Partly due to relative heating of oceans and continents (India, West Africa), brings a season of wet weather. Becomes more prominent when combined with topographic effects that creates a semi-permanent low pressure over land, thereby drawing moisture from the ocean.

4. **Orographic Cooling (lifting):** Cooling as air rises over mountain range. Greater percentage of precipitation occurring on windward side of mountains

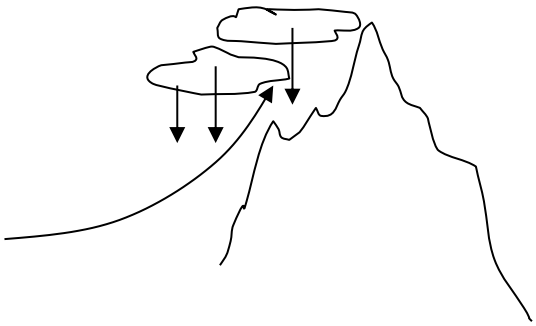


Figure 7: Orographic cooling

B. Change of Phase (nucleation):

Once saturated conditions exist droplets are formed, however surface tension resists nucleation. Therefore, the presence of small particles (aerosols e.g. sea salt, pollutants such as dust and combustion by-products and ice) provides the initial condensation nucleus for droplet formation.

C. Droplet Growth

Principal growth mechanism is coalescence or growth by collision. Collisions are generated by turbulent air motion and falling rain drops. A droplet must grow to a size large enough to overcome the upward convective velocities.

D. Cloud Seeding: Artificial nucleation to induce precipitation.

Rainfall may be induced by the introduction of artificial nucleation materials (such as Silver iodide or dry ice) into the atmosphere. The long term effect of such activity on rainfall variability is not clearly known, as extreme events or events that would have occurred without seeding is difficult to evaluate

Rainfall measurement:

Rain gauges are used to measure the depth and intensity of falling rain on flat surfaces. There are two main types of rain gauges namely: non-recording and recording types.

Non-recording rain gauge: It is used to record the rainfall amount over its duration. It is economical, requiring service after the rains, and is relatively free of maintenance. The measurement is by the use of measuring cylinder or stick. The main sources of error may arise from the measuring stick, leaks in the funnel or can, denting of can, and evaporation.



Figure 8: Standard rain gauge

Recording rain gauge: It is use to record the rainfall amount with time on a continuous basis. It may be a weighing bucket, tipping bucket or float-siphon type. They are suitable for poorly accessible areas and also for determining rainfall intensity.

Weighing Bucket Raingage - weighs the amount of water collected and converts it to mm. The amount is recorded on paper, similar to the way it is done on the thermohygrograph. Newer versions record data in digital form on data loggers.

Tipping bucket - counts the number of tipping per unit time and with the volume of bucket known, the amount of rainfall can be calculated

Float-siphon – float rises with rainfall giving an indication of rainfall amount



Figure 9: Tipping Bucket Raingauge (a) RM Young (0.1mm/tip)

Factors affecting rainfall measurement

1. Design of rain gauge
2. Topography of location
3. Presence of obstruction (such as trees and buildings)

Rainfall data

The important parameters of rainfall are intensity, duration and frequency. The intensity is amount of rainfall per unit time expressed in mm per hour. The time period for a given storm is referred to as the duration. The period taken for a give storm of a certain intensity and duration to re-occur is referred to as the return period or recurrence interval.

Effective rainfall

This is the rainfall that infiltrates the soil and is available for crop use. It is defined as that portion of total rainfall which is directly or indirectly useful for crop production. It must take care of the consumptive use by crops, rainfall for land soaking, land preparation, leaching of salts etc. The effective rainfall values used for irrigation system design and water scheduling varies in different tropical countries (i.e. 70-90 %). The actual effectiveness of rainfall depends on:

1. Rainfall amount and intensity
2. Evaporative demand
3. Land and soil characteristics
4. Antecedent soil-water status
5. Groundwater status
6. Farm and crop management practices
7. Crop characteristics

Temperature

Temperature is the degree of hotness or coldness of a body. For the atmosphere, the air temperature gives how warm the environment is. It is measured with the air thermometer which is placed in the Stevenson screen. In the screen there are 4 thermometers:

1. Dry bulb thermometer for direct reading of air temperature

2. Wet bulb thermometer for reading temperature under moist condition
3. Maximum and minimum thermometers used for reading the maximum and minimum temperature over 24 hour period.

Using the dry and wet bulb readings dew point, vapour pressure, and relative humidity can be estimated.

Air temperature is measured by:

1. Liquid-in-glass thermometer
    a. Mercury-in-glass thermometer for measuring the maximum temperature, T_{max},
    b. Alcohol-in-glass thermometer for measuring the minimum temperature, T_{min},
2. Thermograph: Plots continuous temperature over time for a day or week. The thermograph is place in Stevenson screen.
3. Thermistor: An object whose electrical resistance changes with temperature
4. Thermocouples:

Dry-bulb Temperature: The dry-bulb temperature of air is measured by a thermometer which is freely exposed to the air but is shielded from radiation and moisture.

Wet-bulb Temperature: The Wet-bulb temperature of air is measured by a thermometer whose bulb is covered by a muslin sleeve which is kept moist with distilled clean water, freely exposed to the air and free from radiation.

Maximum thermometer: A maximum thermometer is a mercury-in-glass thermometer made with a constriction in the bore between the bulb and the graduated portion of the stem. Rising temperature forces a portion of the mercury into the graduated section. This mercury is retained and gives a reading of the highest temperature reached until reset.

Minimum thermometer: A minimum thermometer is an alcohol-in-glass thermometer with a freely moving, small, dark-colored glass index placed in the bore. As the temperature falls, the retreating upper end (meniscus) of the alcohol column moves the index toward the bulb. When the temperature rises, the index remains at the lowest point until reset.

- Mean temperature = $(T_{max} + T_{min})/2$
- Temperature range = $T_{max} - T_{min}$

Temperature of a given location – the average daily mean temperature over a period of 30 years

Daily temperature range – the difference between the highest and lowest temperatures recorded in a particular day.

Factors affecting temperature:

1. Latitude,
2. Altitude (the height above mean sea level),
3. Elevation (the height above a given land surface),
4. Wind,
5. Contrast between land and water,
6. Warm and cold ocean currents,
7. Local impact such as slope and orientation.

Seasonal temperature variations are influenced by:

1. Angle of inclination of solar rays
2. Distance between the sun and the earth
3. Air mass movement (i.e. seasonal winds which contribute to rain)
4. Period of the year
5. Land cover type
6. Land topography
7. Prevailing weather situation (cloud cover, moisture, wind)

Importance of temperature to organic life

1. Rate of physical and chemical processes within plants and subsequent control of biological reactions

2. The diffusion rate of gases and liquids varies with temperature
3. Solubility of different organic substances varies with temperature
4. Stability of enzyme and their reactions are affected by temperature

Soil Temperature

Soil temperature plays an important role in many processes, which take place in the soil such as chemical reactions and biological interactions. Soil temperature varies in response to exchange processes that take place primarily through the soil surface. These effects are propagated into the soil profile by transport processes and are influenced by the specific heat capacity, thermal conductivity and thermal diffusivity.

Soil thermometers are specialized thermometers used to measure soil temperatures when place at the desired depth. In modern times some weather stations have automatic soil temperature measurement. Thermistors and thermocouples are some of the standard soil temperature sensors that are available.

Role of soil temperature:

1. Influences soil chemical and biological interactions
2. Seed germination: as optimum soil temperature is required for good germination for any given crop seed
3. Crop yield: soil temperature must reach the optimum for a specific crop to get a good stand
4. Plant growth: soil temperature range optimum for good crop growth of any crop specie
5. Soil temperature influences soil borne diseases like seedling blight, root rot etc.
6. Decomposition of organic matter

Modification of soil thermal regime

Soils' thermal regime can be modified through the use of tillage practices, application of mulch and shelter or shading. Soil thermal regime has great advantages for agricultural practices including controlling germination and plant growth through increasing or decreasing the soils temperature. Mulches, shelters and other soil coverings as well as tillage practices may be used to warm or cool the soil. Mulches, depending on what they are and their albedo, will either warm or cool the soil. A highly reflective material will lower the temperature by reducing the radiant energy reaching the soil. A denser and less reflective material will increase the soil temperature by inhibiting evaporation.

Diurnal variation of soil temperature

Soil temperature increases with depth up to some point, beyond which it remains constant. The value depends on a particular location. Surface temperature is high in the afternoon compared to that in the morning.

Sunshine duration

The amount of sunshine received in a location depends on the latitude and how much cloud is in the sky. The Stokes sunshine recorder made up of spherical glass focuses the sun's rays on to a specially treated calibrated card where a trace is burnt or scorched. The accumulated burnt trace as the sun moves round during the day gives a measure of the total length of bright sunshine hours. The card has to be change daily at the observational time in the morning.

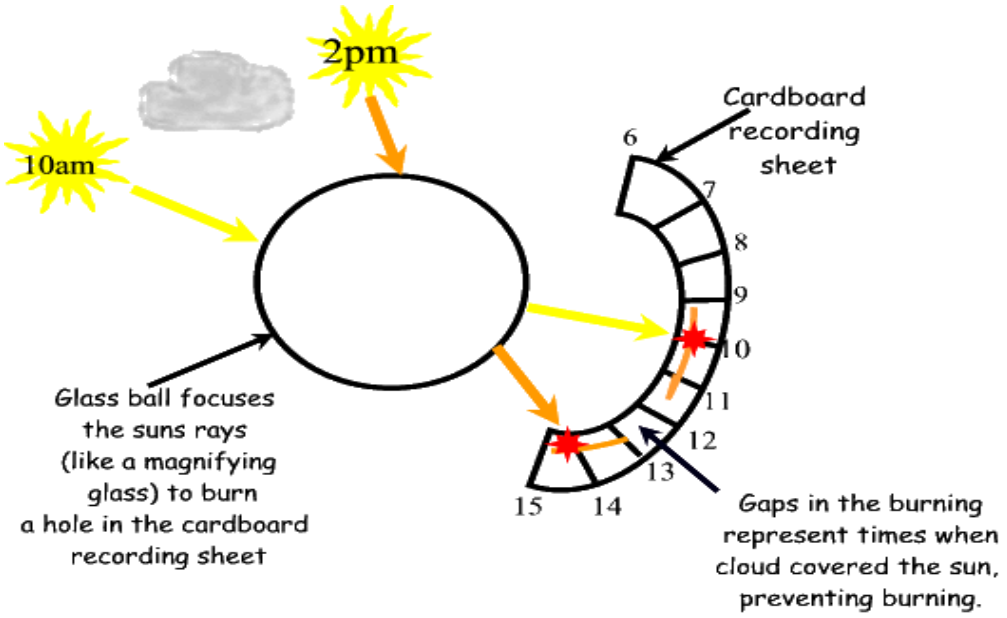


Figure 10: Example of Stokes sunshine recording

Humidity (moisture in the air)

In meteorology, vapour pressure, dew point temperature and relative humidity are common expressions to indicate air humidity.

Vapour pressure

The pressure water vapour exerts if it exists alone at the temperature and pressure of atmospheric air is the vapour pressure. It is the partial pressure of water vapour. Vapour pressure increases with temperature

a. At higher temperature more molecules have the necessary kinetic energy to escape the attractive forces of the liquid phase
b. The more molecules in the vapour phase, the higher the vapour pressure
c. At 100°C the vapour pressure of water is 760 mmHg (1 atm) or equal to the atmospheric pressure of the liquid (in an open container)

d. At high altitudes (i.e. up in the Mountains) the air pressure is less than at sea level. Thus, water will boil at a lower temperature (the vapour pressure needed to support a bubble is lower at high altitude).

e. Water vapour acts as a greenhouse gas by trapping infrared radiation reflected from the earth.

f. Desert temperatures can become much lower at night, as there is little moisture in the air to trap the heat.

Saturated Air

Under a given temperature a fixed volume of air has the capacity to absorb only a certain quantity of water vapour. When this capacity is reached, the air is said to be saturated with water vapour.

a. There is a limit on the amount of vapor the air can hold at a given temperature.

b. Air that is holding as much moisture as it can at a given temperature is called saturated air.

c. Any moisture introduced into saturated air will condense.

Dew Point Temperature

The dew point temperature is the temperature at which the air can no longer hold all of its water vapour, and some of the water vapour must condense into liquid water. Dew point is the temperature at which water vapour saturates from an air mass into liquid or solid usually forming rain, snow, frost, or dew. Dew point normally occurs when a mass of air has a relative humidity of 100%. This happens in the atmosphere as a result of cooling through a number of different processes. Dew point temperature is defined as the temperature to which the air would have to cool (at constant pressure and constant water vapour content) in order to reach saturation.

Dew point depression

It is the difference between the Temperature of the air and the dew point temperature. Low dew point depression values mean that the air is very moist and an increased likelihood of clouds forming. Larger values mean drier air.

Vapour Pressure Deficit/ Saturation deficit

Vapour pressure deficit is the difference between saturation vapour pressure and actual vapour pressure for a given period at the same temperature. It is the amount of additional water vapour that the air can hold. It is an indicator of the evaporative capacity of the air.

Humidity

Absolute Humidity (AH): amount of water vapour in a given volume of air (grams of vapour/cc of air)

Water Vapour Capacity (WVC): amount of water vapour that could be held in a given volume of air. It is dependent on temperature as warm air can hold more moisture than cold air can

Relative Humidity: ratio of how much water the air has to the amount that it could have, expressed in percentage

RH = (AH/WVC)100

$$RH(\%) = \frac{Actual\ vapour\ presure at\ a\ given temp.}{Saturated Vapour\ pressure at\ the\ same temp.} \times 100$$

Characteristics of RH

a. Relative humidity of 100% means the air can hold no more water (rain or dew is likely)
b. A relative humidity of 0% indicates there is no moisture in the atmosphere.
c. Warm air can hold more moisture than cold air. So, if warm air and cold air contain the same amount of moisture, the warm air will have a lower relative humidity.
d. Two conditions cause the relative humidity to rise: when the temperature falls or when moisture is added to the air.
e. Relative humidity, combined with air temperature, can be used to estimate the actual amount of moisture in the atmosphere, sometimes referred to as precipitable water.

Problems or challenges of high relative humidity

a. Mold and condensation problems occur when the relative humidity is too high.

b. The effect of moisture in gases also plays a very significant role in corrosion phenomena which can result in damage and loss of not only unprotected metals, like iron and steel structural components, but also improperly treated or stored steel and other metal products.

Uses of Relative humidity:

a. meteorologists to help predict the weather

b. pathologists to predict disease development on plants, and

c. agricultural scientists to estimate evapotranspiration

Saturation: point at which evaporation is less than condensation

RH = 100%

AH = WVC

Measurement of relative humidity

a. Direct measurement using

i. **Hair hygrometers** which uses human or horse hair as a sensing element. The hair changes in length in proportion to humidity of the air. The response to changes in humidity is slow and is not very dependable at very high relative humidities. The change in length is transmitted and magnified by a lever system into a measurement of relative humidity.

ii. **Electric hygrometers** are based on substances whose electrical properties change as a function of their moisture content. As the humidity of the air around the sensor increases, its moisture content increases, proportionally affecting the sensor's electrical properties. Sensors lose their calibration if allowed to become contaminated or water condenses on them.

b. Indirect measurement - Psychrometer

Psychrometer is a particular kind of hygrometer. It consists of two matched thermometers one of which measures air temperature T_d called the **dry-bulb**. The other thermometer is covered with a wet muslin cloth and reads the **wet-bulb** temperature T_w. Relative humidity can be determined from wet bulb and dry bulb temperatures. The difference in temperatures $T_d - T_w$ is proportional to relative humidity of the air.

Sling Psychrometer

The simplest hygrometer - a sling psychrometer - consists of two thermometers mounted together with a handle attached on a chain. One thermometer is ordinary. The other has a cloth wick over its bulb and is called a wet-bulb thermometer. When a reading is to be taken, the wick is first dipped in water and then the instrument is whirled around. During the whirling, the water evaporates from the wick, cooling the wet-bulb thermometer. Then the temperatures of both thermometers are read. If the surrounding air is dry, more moisture evaporates from the wick, cooling the wet-bulb thermometer more so there is a greater difference between the temperatures of the two thermometers. If the surrounding air is holding as much moisture as possible - if the relative humidity is 100% - there is no difference between the two temperatures. Meteorologists have worked out charts of these differences for each degree of temperature so that the observer can find relative humidity easily. A sample is shown below:

Table 5. Partial Relative Humidity Chart for 30° C Difference Between Dry Bulb and Wet Bulb

Temperatures	**Relative Humidity**
None	100%
0.5	96
1	93
1.5	89
9	44
9.5	42
14.5	19
15	17
18	5

The amount of evaporation, and consequent cooling of the thermometer, depends on the humidity of the atmosphere—the drier the atmosphere, the faster the water evaporates. Using this data and humidity tables or calculations, the dew point (the temperature to which air would have to be cooled for saturation to occur) can be determined, and from it, the relative humidity.

Wet bulb depression – the difference between the dry bulb and the wet bulb temperature readings is called the wet bulb depression.

a. The wet bulb depression increases as the relative humidity decreases
b. The lesser the wet bulb depression, the wetter the air
c. The greater the wet bulb depression the drier the air

Thermohygrograph: The hygrothermograph records both relative humidity and temperature on a graph.

Atmospheric Pressure

It is defined as the force per unit area exerted against a surface by the weight of the air above that surface. Atmospheric pressure is measured with an instrument called a barometer, which is why atmospheric pressure is also referred to as barometric pressure. Meteorologists use millibars (mb),

Pressure decreases with increasing altitude. The number of air molecules above a surface changes as the height of the surface above the ground changes. For example, there are fewer air molecules above the 50 kilometer (km) surface than are found above the 12 km surface. Since the number of air molecules above a surface decreases with height, pressure likewise decreases with height.

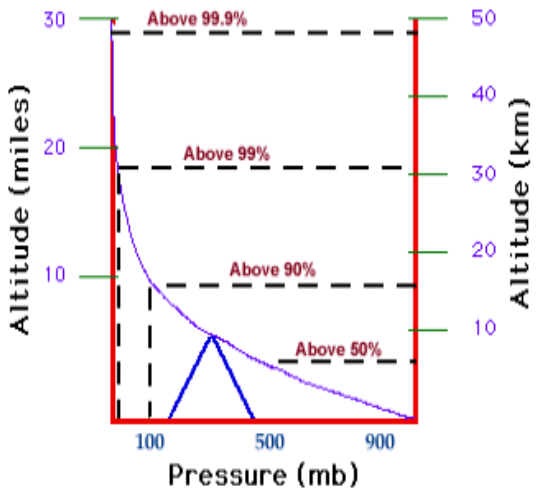


Figure 11: Pressure levels in the atmosphere

Isobars are lines of constant pressure. A line drawn on a weather map connecting points of equal pressure is called an "isobar" and is given in millibars.

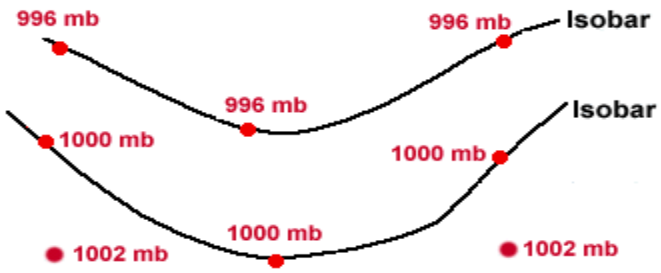


Figure 12: Isobars readings

Wind speed and direction

Wind results from a horizontal difference in air pressure and since the sun heats different parts of the Earth differently, causing pressure differences, the Sun is the driving force for most winds.

Atmospheric pressure is closely determines the weather. Regions of pressure that are slightly higher or slightly lower than the mean atmospheric pressure develop as air circulates around Earth. The air rushes from regions of high pressure to low pressure, causing winds. The properties of the

moving air (cool or warm, dry or humid) will determine the weather for the areas through which it passes. Knowing the location of high and low pressure areas is vital to weather forecasting.

Wind Speed Measurement: Anemometers

Wind is a two-dimensional vector quantity expressed by its speed and direction. The **cup anemometer** fixed on a 2 m pole from the ground is used to record the instantaneous wind speed (m s-1) or the run of the wind (i.e. distance in km). The cup anemometer has a vertical axis and three cups which capture the wind. The number of revolutions per minute is registered electronically.

Wind speed is slowest at the surface, and increases with height until a constant mainstream value is reached at a certain height above the surface. Wind is stronger over the ocean than over land because land surface is rougher than water surfaces. Anemometers are placed at a chosen standard height in an open field location. In meteorology, the standard height is 10 m. Anemometers are located in the field at a point at least 10 times the height of the nearest obstruction. In agrometeorology the standard height is usually 2m. If the wind speed is measured at a height other than 2 m, then the observed values are converted to 2 m height wind speed, using the following relationship:

$$U_{2m} = \frac{4.868}{\ln(67.75\,z - 5.42)} U_z$$

z – the height of wind speed measurement above local ground level

Wind speed may affect:

a. Wind erosion
b. Spread of forest fires and insect pests infestation
c. Transport of water vapour from an evaporating surface
d. Damage to crops and buildings
e. Drive ocean currents

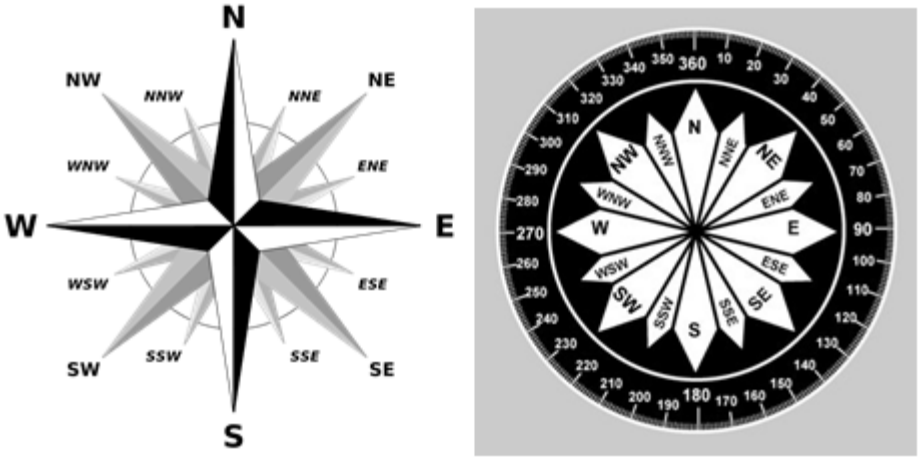
f. Generate surface waves in oceans

Wind direction measurement

Wind direction is the direction from which a wind originates or comes from. It is usually reported in cardinal directions or in azimuth degrees. Normally, the anemometer is fitted with a wind vane to detect the wind direction. Wind vanes point in the direction from which the wind is blowing. A windsock is a conical textile tube designed to indicate wind direction and relative wind speed. Windsocks typically are used at airports and at chemical plants where there is risk of gaseous leakage. They are sometimes located alongside highways at windy locations. Wind direction is the opposite of the direction in which a windsock or windvane is pointing (note that wind directions are conventionally specified as being the compass point from which the wind originates.



Figure 13: Wind vane or windsock



A Compass rose showing the four cardinal directions, the four ordinal directions

A 16-point compass rose/windrose, showing correspondence with 360° circle

Figure 14: Compass rose

Visibility

Meteorological visibility by day is the greatest distance at which a black object of suitable dimension located near the ground can be seen and recognized when observed against a scattering background of fog, sky etc. Meteorological visibility refers to transparency of air. Visibility affects all forms of traffic: road, sailing and aviation.

Radiation measuring instruments

The solar radiation striking a surface is composed of three components. These are:

a. Direct

b. Diffuse

c. Reflected

The generic term for all radiation measuring instruments is radiometer. Radiation measuring instruments can be classified according to their use.

a. **Pyrheliometer** measures the direct solar beam. Its sensing element must be pointed at the sun.

b. **Pyranometer** is used to measure global solar radiation, so it must respond to both the direct solar radiation and to diffuse sky radiation. The sensing element must be a horizontal flat surface

c. **Pyrgeometer** is used to measure global longwave earth radiation

d. **Pyrradiometer** is used to measure total global radiation, both shortwave and longwave. Net pyrradiometer measures the difference between incoming and outgoing radiation

e. **Albedometer** measures the difference between incoming and reflected shortwave radiation (solar) radiation

Evapotranspiration

Evaporation is the transfer of liquid water into the atmosphere. Evaporation occurs when the number of molecules that break from the moist surface and escape into the air as vapour is larger than the number that re-enter the water surface and become trapped. Transpiration is the process by which water vapour passes into the atmosphere through tissues of living plants. Evapotranspiration is the phenomenon of the transfer of water vapour from moist surfaces and living plant into the atmosphere (i.e. evaporation + transpiration). This transfer takes place in two ways.

1. Evaporation from the ground surface
2. Transpiration from the leaf surface of water absorbed by the plants. The amount of water transpired corresponds to a certain amount of production of plant matter.

Evaporative power: is the airs ability to transform a mass of liquid water into vapour and diffuse it in its gaseous mass.

Estimating Evapotranspiration

Methods of predicting evaporation and subsequently evapotranspiration can be put in three groups:

1. **Mass transfer**: Water moves away from evaporating and transpiring surfaces in response to the combined phenomenon of turbulent mixing of the air and vapour pressure gradient. This approach requires the measurement of wind velocity and humidity at two elevations and it is rarely practical
2. **Energy balance**: heat is required for evaporation, therefore if there is no change in water temperature then the net radiation or heat supplied is a measure of evaporation. This approach is more practical. Eg. Penman equation
3. **Empirical methods**: This is based on the assumption that the energy available for evaporation is proportional to the temperature. Eg. Blaney_Craddle

Evaporation from pan: Class A pan is a galvanized pan of diameter of 1.21 m, depth 25.5 cm and 0.8 mm thick. It is painted grey. It is mounted on a wooden stand (15 cm above ground level) such that air freely circulates around and under the pan placed in an open space. A ruler is used to measure the drop in water.

Table 6. Readings from the evaporation pan, in the absence of rainfall

Day/Date	Reading from the pan (one reading every 3 days) mm	Rain (mm)	Average evaporative power per 3 day-period
Monday	0		
Tuesday			
Wednesday			
Thursday	27		27/3=9.0mm/day
Friday			
Saturday		9	
Sunday	25		(25+9)/3= 11.3mm/day
Monday			
Tuesday			
Wednesday	29		29/3=9.7mm/day
Thursday			

Total evaporation for the 9 days is 81m

Average daily Evaporation: (81+9)/9 =10.0 mm/day

In the case of rain: Its depth is added to the reading on the pan gauge e.g. If the reading indicates 23mm and the rainfall recorded was 8mm evaporation during the period amounts to 31mm

Potential evapotranspiration: Is the maximum possible amount of evaporation and transpiration that takes place given unlimited supply of moisture under a given meteorological condition. It is the quantity of water returned to the air by cultivated and uncultivated plants in optimal growing and watering condition. Implying that the soil contains the supply of water needed to satisfy the demands of plants growing.

Read/Actual evapotranspiration: when water supply is not adequate.

Factors that influences evapotranspiration:

- Degree of soil water saturation.
- Soil texture and structure, which influences capillary rise and retention
- Available energy
- Net radiation
- Soil temperature
- Air temperature surrounding the soil surface and the leaves
- Saturation deficit
- Wind speed and roughness of evaporating surface
- Root and leaf development, as it determines the capacity of the plant biomass to take up water in the ground and transpire it into the air
- Level of salt concentration as it decreases with increase in salt concentration of the water
- Elevation: increases with elevation

Reference evapotranspiration (ETo): is the potential evapotranspiration (ET) from a grass reference surface. The reference surface can equally be alfalfa. If the reference surface is a grass, ET stands for the maximum evapotranspiration of an extensive green grass cover of uniform height (8 to 15cm) completely shading the ground, actively growing, and experiencing no shortage of water.

Methods of determining ETo.

- Blaney Criddle.
- Penman ETo according to Frere and Popov.
- Doorenbos and Pruitt.
- Penman-Monteith.

DATA PROCESSING

Rainfall data

The important parameters of rainfall are intensity, duration and frequency. The intensity is the amount of rainfall per unit time expressed in mm per hour. The time period for a given storm is referred to as the duration. The period taken for a give storm of a certain intensity and duration to re-occur is referred to as the return period or recurrence interval.

Rainfall data representation

A. Cumulative mass curve

Typically, the output from a recording gauge is in the form of cumulative rainfall over time. The slope of the cumulative mass curve at any point represents the rate of rainfall (depth/time).

B. Rainfall hyetograph

A *rainfall hyetograph* is a graphical representation of the variation of rainfall depth or intensity with time. Typically the rainfall distribution is represented in discrete time intervals using the cumulative mass curve to calculate the depth over each time interval. The hyetograph is the most common input to hydrologic models (surface hydrology models). Rainfall-runoff hydrograph methods require a description of this variation. It is possible to use actual rain gauge data in rainfall-runoff models if the data are recorded using a small enough time period (such as 15-minute increments). Often such data are not readily available.

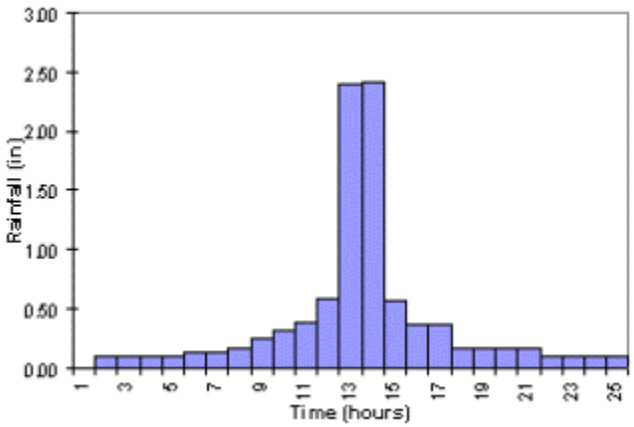


Figure 15: Rainfall hyetograph

C. Isohyetal Maps

Isohyetal map is a plot of constant rainfall depth (sometimes intensity) contours based on rainfall data recorded at gauged locations (points). These are generated similarly to topographic contour maps. Isohyetal maps are used to represent long-term averages or individual storm events.

Point rainfall data assessment

For hydrological works, most often we need to address one or more of the concerns highlighted below when dealing with point rainfall measurements;

1. Estimate average rainfall over large area (watershed)
2. Reduce point rainfall to aerial rainfall for modeling applications to large areas
3. Estimate missing point of gauge data or estimate data at an ungauged location

Area precipitation: The objective is to estimate an average measure of rainfall over an area or region (sometimes on an event basis or usually for annual averages).

1. *Arithmetic Averages*: In using this approach one has to have
   - gauges that are uniformly distributed
   - individual gauge measurements that do not vary greatly (typically smaller areas)

$$Pavg = \frac{\sum_{i=1}^{n} P_i}{n}$$

P_i = rainfall at gauge i

n = number of gauges

2. *Thiessen Polygon*: The underlying concept is that rainfall for an area is represented by the nearest gauge. The procedure is:
   a. draw lines to connect each gauge
   b. draw perpendicular bisector on each connecting line

c. connect bisectors to form polygons

$$Pavg = \frac{\sum_{i=1}^{n} A_i P_i}{A_T}$$

n = number of polygons (gauges)

A_T = total area

A_i = area of polygon around gauge i

P_i = rainfall at gauge i

Note if gauge location changes polygons will change also.

3. Isohyetal Method: Proceedure

- construct isohyetals between gauges
- calculate area between isohyets
- estimate average rainfall between isohyets applying the same summation as the Thiessen polygon where A_i and P_i represent the area and average rain between two isohyetals.

Note that isohyets should remain unchanged for long-term even with change in gauge location

Missing data points or rainfall at ungauged locations

This is usually done for a long term record. There are three methods commonly used, that are based on long term records of surrounding gauges – require a minimum of three.

1. Simple average of surrounding gauges
   - Gauges must be equal distance from point of estimate
   - They should have fairly equal rainfall at known gauges

2. Normal Ratio Method for missing data estimation
   - when surrounding gauges differ by greater than 10% in measured rainfall
   - stations are weighted based on normal annual precipitation ratio's

   Example when estimating rainfall for location x

$$P_x = \frac{1}{k}\left(\frac{N_x}{N_1}P_1 + \frac{N_x}{N_2}P_2 + ... + \frac{N_x}{N_k}P_k\right)$$

where: N_x = normal long-term annual precipitation at location x

P_k = event precipitation at gauge k

N_k = normal long-term annual precipitation at gauge k

P_x = estimated event precipitation at location x

The event might be a single storm total, daily total, monthly total or annual value.

3. Reciprocal (Inverse weighted) Distance Method: This is used for missing data and ungauged location data estimation. Surrounding gauges are weighted based on reciprocals of the sum of squares of distance away from the point of interest.

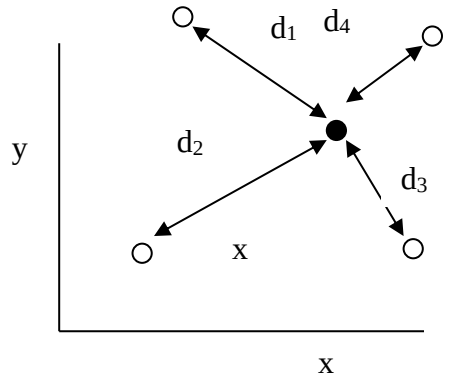


Figure 16: Inverse Weighted Distance method

X and y are distances from the ungauged point (missing data) or location where precipitation is being estimated

$$d_i = \sqrt{x^2 + y^2}$$

$$W_i = \frac{1}{d_i^2}$$

$$P_x = \frac{\sum_{i=1}^{n} P_i W_i}{\sum_{i=1}^{n} W_i}$$

CLIMATE CHANGE

Climate change is a long-term change in the statistical distribution of weather patterns over periods of time. This may be a change in the average weather conditions or a change in the distribution of weather events with respect to an average, for example, greater or fewer extreme weather events. Climate change may be limited to a specific region, or may occur across the whole Earth.

Global climate change is caused by the accumulation or increase in the concentration of greenhouse gases (water vapor, CO_2, CH_4, N_2O, and CFCs) in the lower atmosphere, which trap infrared radiation from the Earth's surface and thus cause the greenhouse effect. This effect is a natural phenomenon, which helps maintain a stable temperature and climate on Earth. Human activities, such as coal burning, fossil fuel combustion (which release carbon dioxide), deforestation (because forests remove carbon from the atmosphere), methane gas released in animal waste and some industrial processes have led to an increase in greenhouse gases concentration. Consequently, more infrared radiation has been captured in the atmosphere, which causes changes in the air temperature, precipitation patterns, sea-level rise, and melting of glaciers. Projections of future climate change are derived from global climate model or general circulation model (GCM) experiments. It is estimated that global mean surface temperature will rise by 1.5° to 3.5° C by 2100. Large changes in precipitation, both increases and decreases, are forecast, largely in the tropics. Climate change is very likely to affect the frequency and intensity of weather events, such as storms and floods around the world. Climate change will also cause sea level rise due to the thermal expansion of the oceans and the melting of the mountain glaciers. Global mean sea level is anticipated to rise by 15 to 95 centimeters by 2100. Sea level rise will increase vulnerability to coastal flooding and storm surges. Many ecosystems may decline or fragment and individual species may become extinct.

Global warming

Global warming is the gradual increase of the temperature of the earth's lower atmosphere and oceans and its projected continuation as a result of the increase in greenhouse gases since the Industrial Revolution. The temperature of the atmosphere near the earth's surface is warmed through a natural process called the greenhouse effect. Visible, shortwave light comes from the sun to the earth, passing unimpeded through a blanket of thermal, or greenhouse gases composed

largely of water vapor, carbon dioxide, methane, nitrous oxide, and ozone. Infrared radiation reflects off the planet's surface toward space but does not easily pass through the thermal blanket. Some of it is trapped and reflected downward, keeping the planet at an average temperature suitable to life, about 16°C.

Growth in industry, agriculture, and transportation since the Industrial Revolution has produced additional quantities of the natural greenhouse gases plus smaller quantities of chlorofluorocarbons and other more potent greenhouse gases, augmenting the thermal blanket. It is generally accepted that this increase in the quantity of greenhouse gases is trapping more heat and increasing global temperatures, making a process that has been beneficial to life potentially disruptive and harmful. Some projected, longer-term results of global warming include melting of polar ice, with a resulting rise in sea level and coastal flooding; profound changes in agriculture due to climate change; extinction of species as ecological niches disappear; more frequent tropical storms; expansion of subtropical deserts; and an increased incidence of tropical diseases. As a result of contemporary increases in atmospheric carbon dioxide, the oceans have become more acidic; a result that is predicted to continue.

Greenhouse Gas (GHG)

A **greenhouse gas** is any gas in the atmosphere that absorbs and emits radiation within the thermal infrared range. The Earth's atmosphere, thus acts like the glass of a green house, hence the 'greenhouse effect'. This process is the fundamental cause of the greenhouse effect. The primary greenhouse gases in the Earth's atmosphere are water vapor, carbon dioxide, methane, nitrous oxide, and ozone. Greenhouse gases greatly affect the temperature of the Earth; without them, Earth's surface would be on the average about 33 °C colder than at present.

Table 7. Levels of GHG at the pre-industrial and current

Gas	**Pre-industrial level**	**Current level**	**Increase since 1750**	**Radiative forcing (W/m^2)**
Carbon dioxide	280 ppm	388 ppm	108 ppm	1.46
Methane	700 ppb	1745 ppb	1045 ppb	0.48
Nitrous oxide	270 ppb	314 ppb	44 ppb	0.15
CFC-12	0	533 ppt	533 ppt	0.17

Climate change impact

Table 8. Climate change related risk and impacts

Sectors	Climate-related risks	Impacts
Health	Heatwave/ coldwave	Cardiovascular, respiratory and heatstroke mortality
	Flood, landslide, windstorm	Death and injuries, infectious diseases and mental disorders
	Drought	Starvation, malnutrition and diarrhea and respiratory diseases; strain on health due to poor drinking water quality and availability
	Extreme temperature and excess of rainfall	Mosquito-borne and tick-borne diseases; rodent-borne, waterborne and food-borne diseases
Agriculture	Flood, heavy rainfall, hailstorm	Effect on seedlings, damage to crops and submergence, inefficiency of applied fertilizers; damage to food and shelter for livestock; disease such as cholera, worm infestation
	Drought	Crop establishment in high lands, low plant stand, damage to crops; outbreak of diseases such as black quarter, anthrax in cattle
	High/ low temperature	Yield reduction; diseases such as foot-and-mouth diseases, peste des petits in cattle, heatstroke and production loss, cold stress and production loss
Tourism	High/ low temperature	Travelers may be subject to sudden changes in temperature thus the body requiring time to adjust
	Radiation	Harmful to skin and eyes
	Wind	Adds to the discomfort of people, especially when laden with moisture
	High humidity and temperature	May lead to dehydration and even fatality
Transport	Flood, heavy precipitation	Flooding of roadways, rail lines, subterranean tunnels and runways, road washout, damage to rail-bed support structures, damage to pipelines
	Heat wave	Compromised pavement integrity, deformation of rail lines, thermal expansion of bridge joints, heat buckling or runways
	Drought	Increased risk of wildfires, decreased visibility at airports in drought-prone areas
Water resources	Heavy rainfall	Increased river discharge, inundation, dam management

	Dry spell	Poor water quality, reduced water resources, effect on reservoir management and freshwater distribution in urban areas
Energy	Heat wave / cold wave	Increased demand for heating or cooling, reduced energy supply, affects gas and fuel pipelines
	Precipitation deficiency	Reduced hydropower energy production

Climate change effect on agriculture in the tropics

The responses of living organisms are the best indicators of climate change. Micro-organisms are the first to respond to a changing climate. Insects and short-lived plants are next. Probably amphibians will respond next. These organisms will move as the climate changes and their range will increase or decrease. As the climate changes, some organisms die, while others will begin to thrive. Climate change has many important implications in environmental science. When climate changes, some species may become extinct if it cannot move with the changing climate.

With climate change the agricultural engineer has to develop and adapt machinery for crop production and storage. Also the engineer has to adapt strategies as it relates to soil and water management as crop water requirements, rainfall amounts and patterns, will be affected.

Soil processes: The potential for soils to support agriculture and distribution of land use will be influenced by changes in soil water balance: Increase in soil water deficits i.e. dry soils become drier, therefore increasing the need for irrigation.

Crops: The effect of increased temperature and CO_2 levels on arable crops will be broadly neutral: New crop varieties may need to be selected. Horticultural crops are more susceptible to changing conditions than arable crops. Field vegetables will be particularly affected by temperature changes, water deficits will directly affect fruit and vegetable production. Increase in disease transmission by faster growth rates of pathogens in the environment and more efficient and abundant vectors (such as insects) will have consequences for food quality and storage.

Livestock and poultry: Livestock and poultry could be exposed to higher incidences of heat stress, thus influencing productivity.

Weeds, pests and diseases: Weeds evolve rapidly to overcome control measures, short lived weeds and those that spread vegetatively (creeping buttercup, couch etc) evolve at the greatest rate:

Responses to global warming

There are two basic approaches to limiting climate change or its impact. These are mitigation or adaptation strategies.

Mitigation

Reducing the amount of future climate change is called mitigation of climate change. The IPCC defines mitigation as activities that reduce greenhouse gas (GHG) emissions, or enhance the capacity of carbon sinks to absorb GHGs from the atmosphere. Many countries, both developing and developed, are aiming to use cleaner and less polluting technologies to aid mitigation and could result in substantial reductions in CO_2 emissions. Policies on mitigation include targets for emissions reductions, increased use of renewable energy, and increased energy efficiency.

Adaptation

The other policy response include adaptation to climate change. Adaptation to climate change may be planned, e.g., by local, national government or private, or spontaneous intervention aim at reducing the impact of climate change. The ability to adapt is closely linked to social and economic development. Even societies with high capacities to adapt are still vulnerable to climate change. Planned adaptation is already occurring on a limited basis.